CSE0613316 — Software Project Design and Development (Lab)

SOFTWARE TEST REPORT

Test Plan · Test Specification · Test Completion Report

Tailored from ISO/IEC/IEEE 29119-3

Project title	HeritageBD – Digital Heritage Tourism & Marketplace Platform of Bangladesh
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Experiment	Exp 9 (v1.0) ☑ Exp 10 (v1.1) ☐
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Document Control

Repository URL	https://github.com/Monika-Akter47/heritagebd
Version tested (tag / commit SHA)	v1.0.1-test / 1d45c59 — all results in this report come from this version
Test environment URL (if deployed)	https://heritagebd-flax.vercel.app (API: https://heritagebd-api.onrender.com)
Report version	1.0 (Exp 9)
Prepared by	Monika Akter, Sumaiya Jahan (test suite written with Claude Code; see §10)
Reviewed by (other member)	____
Target tier (Guideline §3)	Baseline ☑ Professional ☐ Distinction ☐

Revision history

Version	Date	Author	Changes
1.0	26 Oct 2026	Team ____	Initial report for Experiment 9: test plan, 163 automated test cases, results, coverage, non-functional checks, defect log
1.1			Planned: static analysis (SonarCloud) and OWASP ZAP results (Experiment 10)

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1. Introduction

1.1 Purpose

This report covers the testing of HeritageBD at version v1.0.1-test (1d45c59): the test strategy, the test cases and how they trace to the SRS, the execution results, quality measurements (coverage, performance, security, accessibility) and the release recommendation. Every number is produced by an automated run or a tool and is listed in Appendix A.

1.2 Product overview

HeritageBD is a web platform for heritage tourism in Bangladesh. Tourists discover heritage places, local food and products, book verified tour guides, plan trips, buy products for delivery and pre-order products for pickup at a destination ("Ready When You Arrive"); vendors, guides and administrators manage their side of each flow. The system is a React 19 + Vite single-page application (Vercel), a PHP 8.2 REST API (Render, Docker) and a PostgreSQL database (Supabase); MySQL is supported for local development.

1.3 Scope of testing

In scope

Feature / module	Requirement IDs	Test levels applied
User management & authentication	FR-01 – FR-06, FR-48, NFR-04 – NFR-06	Unit, Integration, E2E, Security
Heritage discovery, content, categories	FR-07 – FR-10	Unit, Integration, E2E
Interactive map	FR-11 – FR-13, NFR-09	Integration, E2E
Tour guides & bookings, guide payments	FR-14 – FR-17	Integration
Trip planner	FR-18, FR-19	Integration
Marketplace, cart, checkout, delivery orders	FR-20 – FR-24, FR-36, FR-38	Unit, Integration
Pre-orders "Ready When You Arrive" & payments	FR-25 – FR-35, FR-37, NFR-10, NFR-11	Unit, Integration, E2E
Reviews, notifications, chat	FR-39 – FR-43	Integration
Administration & reporting	FR-44 – FR-47	Integration, E2E
Security & reliability requirements	FR-49 – FR-51, NFR-03, NFR-07, NFR-08	Integration, Security
Performance, accessibility, compatibility	NFR-01, NFR-02, NFR-12 – NFR-15	Performance, Accessibility, Compatibility

Out of scope

Feature / item	Reason not tested
Real payment gateways (bKash, Nagad, card)	Not integrated (SRS §6.5): payments are recorded manually with payer number and TrxID; only that recording is tested
E-mail / SMS delivery of the reset OTP	No mail provider configured; OTP handling is tested by inserting a known code into the test database (TC-13)
Supabase / Render / Vercel infrastructure	Owned by the providers; only our use of them is tested (connection, HTTPS, deploy behaviour)
Firefox and Safari	Firefox could not run on the test machine (blocked by Windows application control); Safari needs macOS/iOS. Listed as residual risk §11.2

1.4 References

Document	Version / link
SRS (Experiment 2)	HeritageBD_SRS_After.pdf, 14/10/2026 (51 functional requirements FR-01 – FR-51)
Design document (Experiment 3)	docs/architecture.md, docs/database.md, docs/diagrams.md in the repository

Document	Version / link
Software Testing Report Guideline	CSE0613316 Testing Report Guideline (UITS)
ISO/IEC/IEEE 29119-3; ISO/IEC 25010; ISTQB CTFL; OWASP Top 10 & ASVS L1; WCAG 2.2 AA	—

1.5 Definitions and abbreviations

RTM — Requirements Traceability Matrix; EP — Equivalence Partitioning; BVA — Boundary Value Analysis; E2E — End-to-End; p95 — 95th-percentile response time; TC — test case; TrxID — mobile-wallet transaction ID; PGlite — PostgreSQL compiled to WebAssembly, used as the throwaway test database; S1–S4 — defect severity; P1–P3 — priority.

2. Test Environment

2.1 Environment configuration

Operating system(s)	Windows 11 Pro 10.0.26200 (test machine); production containers on Linux (Render, Vercel)
Runtime / language version	PHP 8.2.35 (NTS, with pdo_pgsql, curl, mbstring; Xdebug 3.5.3 for coverage); Node.js 24.20 (project pins Node 22 in .mise.toml)
Framework(s)	React 19, Vite 8, TypeScript 5.7, Tailwind CSS 4; plain PHP 8 REST API with PDO
Database (test instance)	PostgreSQL 17 in-process (PGlite 0.5) started by tests/support/test-db.mjs, loaded from backend/database/postgres/schema.sql + seed.sql on every run
Browsers / devices	Microsoft Edge 155 (desktop), Edge with Pixel 7 emulation (mobile viewport); Lighthouse on Edge headless
Test hosting	Local: PHP built-in server + Vite dev server on random localhost ports; performance and Lighthouse against the deployed site
Test tools and versions	PHPUnit 11.5.57, phpunit/php-code-coverage (Xdebug), Vitest 4.1.12 (+ @vitest/coverage-v8), Playwright 1.56.1, Lighthouse 12.8.2, npm audit (npm 11.19), composer audit 2.8.12, gitleaks 8.30.1

2.2 Test data

Every run starts a new in-memory PostgreSQL database loaded from the committed schema and demo seed (backend/database/postgres), so runs are independent and leave nothing behind. Tests create their own records (accounts, vendors, guides, products, orders) with random suffixes, so they do not depend on each other or on run order; a few tests arrange state directly in the test database (e.g. an expired token, a known OTP hash). Seed accounts are fictitious demo users (example.com addresses) with a fixture password that exists only in the test database. No real personal data or production credentials are used; the production database is never touched by the functional tests.

2.3 How to run the tests

These commands are the same as README §9 and were verified on a fresh clone of the tag (Appendix A, A9).

```
git clone https://github.com/Monika-Akter47/heritagebd.git && cd heritagebd && git checkout
v1.0.1-test
npm ci
composer install
npm run lint            # tsc --noEmit + php -l on every PHP file
npm test                # PHPUnit unit + API integration, then Vitest
npm run test:e2e        # Playwright journeys (Edge + mobile viewport)
npm run test:coverage   # as npm test, with coverage reports
node tests/performance/p95.mjs   # p95 of key endpoints (deployed site)
```

3. Test Strategy

3.1 Test levels and types

Level / type	Objective	Technique(s)	Tool	Auto?	Owner
Unit	Business rules in isolation: money split, phone/TrxID validation, area resolution, helpers, UI validators	EP, BVA, decision table	PHPUnit, Vitest	Y	Team
Integration / API	Each endpoint with the real router, controllers and a PostgreSQL database, over HTTP	EP, BVA, state transition, error guessing	PHPUnit + curl, PGlite	Y	Team
System / E2E	Critical journeys through the real UI	Use-case scenarios	Playwright	Y	Team
Acceptance (UAT)	SRS acceptance criteria per FR (expected values taken from the SRS)	User-story criteria	PHPUnit / Playwright	Y	Team
Regression	Re-run all suites after each change; one test per fixed defect	Suite re-run	npm test	Y	Team
Performance	p95 response time of 5 key read endpoints	Load (60 req × 5 concurrent)	tests/performance/p95.mjs	Y	Team
Security	Access control, injection, auth, misconfiguration, dependencies, secrets	OWASP Top 10 checklist	PHPUnit, npm/composer audit, gitleaks, curl	Y	Team
Accessibility	WCAG 2.2 AA checks on key screens	Automated audit	Lighthouse	Y	Team
Compatibility	Desktop and mobile viewport	E2E per browser project	Playwright (Edge)	Y	Team
Usability	Core tasks by outside users	Moderated task test	—	N	Team

Tool choice. The guideline's PHP row suggests PHPUnit; the API is plain PHP (no Laravel), so PHPUnit is used for unit tests and for HTTP-level integration tests against the real API, which also exercises routing and SQL on PostgreSQL. Vitest and Playwright are the guideline's JavaScript/TypeScript tools. Substitution: instead of a Dockerised database, PGlite runs PostgreSQL in-process, so the suite needs no Docker or database server and resets on every run. Coverage of the API, which runs in a separate PHP server process, is collected with Xdebug per request and merged (tests/support/merge-coverage.php).

3.2 Entry criteria

- ☑ Feature merged to main and builds without errors (npm run build, npm run lint: 0 errors)
- ☑ SRS acceptance criteria exist for every requirement under test (SRS §5.7 FR-01 – FR-51; NFRs from §5.1 – §5.4)
- ☑ Test environment and test data ready and documented (§2)
- ☑ Dependencies installed reproducibly from lockfiles (package-lock.json, pnpm-lock.yaml, composer.lock)

3.3 Exit criteria

Criterion	Target (Baseline)	Achieved	Met?	Evidence
FRs with ≥1 test case	100%	51/51 (100%)	Met	A3 RTM
Priority-1 test cases passing	100%	96/96 (100%)	Met	A1, A2
Line coverage (core layer)	≥ 60%	92.7% business logic; 78.4% controllers; 80.2% overall	Met	A4
Branch coverage (core layer)	n/a at Baseline	Not measured (Xdebug line mode)	—	—

Criterion	Target (Baseline)	Achieved	Met?	Evidence
S1 / S2 defects still open	0 / (none required)	0 / 1	Met	§9
Static analysis: linter errors	0	0 (tsc --noEmit; php -l on 53 files)	Met	A10
Dependency audit: Critical / High	0 unaddressed	npm 0/0; Composer 0/0	Met	A6
Secret scan	Clean	0 leaks (git history + working tree)	Met	A6
p95 response time (3+ key endpoints)	Measured; team target < 2 s	places 862 ms; products?limit=50 1103 ms; guides 838 ms; places/p1 473 ms; events?upcoming=1 389 ms	Met	A5
Lighthouse accessibility (key screens)	≥ 80	82–95 on 6 pages	Met	A7
Browsers tested	2	1 engine (Edge desktop + mobile viewport)	Not met	A8; §11.2
Usability: 3 outside users	3 users	Not conducted yet	Not met	§8.5
All unit + integration tests run with one command	Yes	npm test (verified on a fresh clone)	Met	A9

4. Requirements Traceability Matrix

Generated from the tests themselves (tests/support/rtm.mjs): every test carries a tag "TC-ID | requirement | level | technique | priority | ±", and the status column is the JUnit result of the tagged run. Full list in evidence/rtm.csv. Status: Pass = all linked cases pass; Partial = some fail; Fail = all fail. NFRs are stated in SRS §5.1 – §5.4 without IDs and are numbered NFR-01 – NFR-16 here.

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
FR-01	User registration	High	TC-01, TC-02, TC-03, TC-04, TC-05 … (7)	I	Pass
FR-02	User authentication	High	TC-U20, TC-07, TC-08, TC-09, TC-10, TC-116	U, I	Pass
FR-03	Role-based access control	High	TC-21, TC-114	I	Pass
FR-04	Vendor and guide verification	High	TC-28, TC-29, TC-30, TC-31	I	Pass
FR-05	Password reset (6-digit OTP, 15 min)	Med	TC-13, TC-14, TC-15, TC-16	I	Pass
FR-06	Profile management, saved places	Med	TC-17, TC-18, TC-19, TC-121, TC-122	I	Pass
FR-07	Browsing and search with filters	Med	TC-123, TC-40, TC-41, TC-42, TC-43, TC-44	I	Pass
FR-08	Place details	Med	TC-45, TC-46	I	Pass
FR-09	Content management (admin)	Med	TC-37, TC-38, TC-39, TC-124, TC-125	I	Pass
FR-10	Categories with counts	Med	TC-47	I	Pass
FR-11	Map rendering	Med	TC-48, E2E-01	I, E2E	Pass
FR-12	Map layer filtering	Med	E2E-01	E2E	Pass
FR-13	Marker detail cards	Med	E2E-01	E2E	Pass
FR-14	Guide profile	Med	TC-52	I	Pass
FR-15	Availability & location sharing	Med	TC-U19, TC-53, TC-54	U, I	Pass
FR-16	Booking requests	High	TC-55, TC-56, TC-57, TC-58, TC-108 … (8)	I, U	Pass
FR-17	Guide dashboard	Med	TC-59	I	Pass
FR-18	Trip creation	Med	TC-60, TC-61	I	Pass
FR-19	Itinerary management	Med	TC-62, TC-63	I	Pass

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
FR-20	Product listing (vendor)	Med	TC-U16, TC-U17, TC-U18, TC-118, TC-119 … (7)	U, I	Pass
FR-21	Product browsing	Med	TC-U14, TC-U15, TC-49, TC-50, TC-51	U, I	Pass
FR-22	Cart	Med	TC-66, TC-67	I	Pass
FR-23	Checkout	High	TC-68, TC-69, TC-F08	I, U	Pass
FR-24	Inventory validation	High	TC-70, TC-71	I	Pass
FR-25	Pre-order eligibility	Med	TC-76, TC-77	I	Pass
FR-26	Destination-based discovery	Med	TC-U10, TC-U11, TC-U12, TC-U13, TC-78	U, I	Pass
FR-27	Pickup scheduling	Med	TC-79, E2E-02	I, E2E	Pass
FR-28	Advance payment calculation	High	TC-U01, TC-U02, TC-U03, TC-U04, TC-U05 … (9)	U, I, E2E	Pass
FR-29	Advance payment processing	High	TC-U08, TC-U09, TC-81, TC-82, TC-83 … (11)	U, I, E2E	Pass
FR-30	Order number and QR code	High	TC-U21, TC-84, E2E-02	U, I, E2E	Pass
FR-31	Pre-order status updates	Med	TC-85, TC-86	I	Pass
FR-32	Pickup verification (QR, single use)	High	TC-87, TC-88	I	Pass
FR-33	Remaining payment	High	TC-89, TC-90, TC-F05, TC-F07	I, U	Pass
FR-34	Cancellation & refunds	High	TC-91, TC-92, TC-93, TC-126, TC-94	I	Pass
FR-35	Pre-order settings	Med	TC-95, TC-96	I	Pass
FR-36	Delivery order tracking	Med	TC-72, TC-73, TC-74, TC-F10	I, U	Partial
FR-37	Pre-order tracking	Med	TC-97, TC-F09, TC-F11, TC-F13	I, U	Pass
FR-38	Order history (role-aware)	Med	TC-75	I	Pass
FR-39	Review submission	Med	TC-98, TC-99, TC-100, TC-101	I	Partial
FR-40	Aggregate rating	Med	TC-102	I	Pass
FR-41	Review display	Med	TC-103	I	Pass
FR-42	Notifications	Med	TC-104, TC-105	I	Pass
FR-43	Chat	Med	TC-106, TC-107	I	Pass
FR-44	User management (admin)	Med	TC-32, TC-33	I	Pass
FR-45	Pre-order management (admin)	Med	TC-34, TC-111, E2E-03	I, E2E	Pass
FR-46	Statistics	Med	TC-35	I	Pass
FR-47	Reports	Med	TC-36	I	Pass
FR-48	Secure login / session expiry	Med	TC-11, TC-12	I	Pass
FR-49	Unauthorized access prevention (401/403)	High	TC-20, TC-22, TC-23, TC-24, TC-25 … (8)	I	Pass
FR-50	Activity logging	Med	TC-112, TC-113	I	Partial
FR-51	Backup and recovery	Med	TC-117	I	Pass
NFR-01	Fast response under normal load (SRS §5.1); team target p95 < 2 s	Med	PT-01, PT-02, PT-03, PT-04, PT-05	Perf	Pass
NFR-02	List endpoints paginated, limit capped at 200 (§5.1)	Med	TC-43	I	Pass
NFR-03	Multi-table writes in transactions with rollback (§5.2)	High	TC-70, TC-71	I	Pass
NFR-04	Passwords stored only as bcrypt hashes (§5.3)	High	TC-116, TC-U20	U, I	Pass

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
NFR-05	256-bit tokens, 30-day expiry, revoked on logout (§5.3)	High	TC-01, TC-11, TC-12	I	Pass
NFR-06	Role guards on protected endpoints; suspended users cannot log in (§5.3)	High	TC-20, TC-21, TC-10, TC-33	I	Pass
NFR-07	PDO prepared statements prevent SQL injection (§5.3)	High	TC-114	I, Sec	Pass
NFR-08	HTTPS in production; CORS restricted to the frontend (§5.3)	High	SEC-01, SEC-02	Sec	Partial
NFR-09	Explicit, revocable guide location sharing (§5.3)	High	TC-53, TC-54	I	Pass
NFR-10	QR codes unique, single-use, validated server-side (§5.3)	High	TC-84, TC-87, TC-88	I	Pass
NFR-11	No unauthorized change of pre-order status or payment records (§5.3)	High	TC-27, TC-126, TC-110	I	Pass
NFR-12	Responsive on desktop, tablet and mobile (§3.1, §5.4)	Med	E2E-01, E2E-02, E2E-03	E2E	Pass
NFR-13	Bangla-capable storage and display (§5.4)	Med	TC-123, TC-F07	I, U	Pass
NFR-14	Accessible user interface (WCAG 2.2 AA, Guideline §3)	Med	LH-01, LH-02, LH-03, LH-04, LH-05, LH-06	A11y	Pass
NFR-15	Modern browsers: Chrome, Firefox, Safari, Edge (§2.4)	Med	E2E-01, E2E-02, E2E-03, FF-01	Compat	Partial
NFR-16	Regular database backups and recovery (§5.2)	Med	TC-117	I	Pass

Total requirements	67 (51 FR + 16 NFR)
Requirements with ≥1 test	67 / 67 (100%)
Requirements with ≥1 negative test	41 / 67 (61.2%)

5. Test Case Specification

5.1 Test case summary

163 test cases (239 executions, counting data-provider rows and browser projects). Source: TA = tool-assisted (see §10). Result counts passed/total executions where a case runs several times.

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-01	FR-01	Tourist registration is active immediately and returns token	Integ	EP	P1	+	Y	TA	Pass
TC-02	FR-01	Vendor and guide registrations start pending verification	Integ	EP	P1	+	Y	TA	Pass
TC-03	FR-01	Password shorter than 8 characters is rejected	Integ	BVA	P1	−	Y	TA	Pass (4/4)
TC-04	FR-01	Second account with same email is rejected case insensitively	Integ	EP	P1	−	Y	TA	Pass
TC-05	FR-01	Admin role cannot self register	Integ	EP	P1	−	Y	TA	Pass
TC-06	FR-01	Invalid registration input is rejected	Integ	EP	P2	−	Y	TA	Pass (4/4)
TC-07	FR-02	Valid login returns user and token usable on me	Integ	EP	P1	+	Y	TA	Pass
TC-08	FR-02	Wrong password and unknown email get the same 401	Integ	EP	P1	−	Y	TA	Pass
TC-09	FR-02	Sixth failed login within 15 minutes is throttled	Integ	State transition	P1	−	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-10	FR-02	Suspended account cannot log in	Integ	State transition	P1	−	Y	TA	Pass
TC-11	FR-48	Logout revokes the token	Integ	State transition	P1	−	Y	TA	Pass
TC-12	FR-48	Expired token is rejected	Integ	BVA	P1	−	Y	TA	Pass
TC-13	FR-05	Valid otp resets password and signs out old sessions	Integ	EP	P1	+	Y	TA	Pass
TC-14	FR-05	Otp older than 15 minutes is rejected	Integ	BVA	P1	−	Y	TA	Pass
TC-15	FR-05	Otp must be exactly six digits	Integ	EP	P2	−	Y	TA	Pass (3/3)
TC-16	FR-05	Forgot password answers the same for unknown email	Integ	EP	P2	+	Y	TA	Pass
TC-17	FR-06	User updates own name and phone	Integ	EP	P2	+	Y	TA	Pass
TC-18	FR-06	Tourist saves and unsaves a place	Integ	State transition	P2	+	Y	TA	Pass
TC-19	FR-06	Wrong current password cannot change password	Integ	EP	P2	−	Y	TA	Pass
TC-20	FR-49	Protected endpoints return 401 without a token	Integ	EP	P1	−	Y	TA	Pass (10/10)
TC-21	FR-03	Each role is refused endpoints of other roles	Integ	Decision table	P1	−	Y	TA	Pass (7/7)
TC-22	FR-49	Tourist cannot read another tourists preorder	Integ	Error guessing	P1	−	Y	TA	Pass
TC-23	FR-49	Tourist cannot read another tourists order	Integ	Error guessing	P1	−	Y	TA	Pass
TC-24	FR-49	Tourist cannot read another users conversation	Integ	Error guessing	P1	−	Y	TA	Pass
TC-25	FR-49	Guide cannot edit another guides profile	Integ	Error guessing	P1	−	Y	TA	Pass
TC-26	FR-49	Vendor cannot edit another vendors product	Integ	Error guessing	P1	−	Y	TA	Pass
TC-27	FR-49	Tourist cannot advance their own preorder to vendor only states	Integ	Error guessing	P1	−	Y	TA	Pass
TC-28	FR-04	Approving a pending guide makes it verified active and notifies it	Integ	State transition	P1	+	Y	TA	Pass
TC-29	FR-04	Rejecting a pending vendor suspends the account	Integ	State transition	P1	−	Y	TA	Pass
TC-30	FR-04	Verification status must be verified or rejected	Integ	EP	P2	−	Y	TA	Pass
TC-31	FR-04	Only an admin can verify	Integ	EP	P1	−	Y	TA	Pass
TC-32	FR-44	Admin lists users filtered by role	Integ	EP	P2	+	Y	TA	Pass
TC-33	FR-44	Admin suspends and reinstates a user	Integ	State transition	P1	+	Y	TA	Pass
TC-34	FR-45	Admin sees preorders of every tourist	Integ	EP	P2	+	Y	TA	Pass
TC-35	FR-46	Statistics counters match the database	Integ	EP	P1	+	Y	TA	Pass
TC-36	FR-47	Reports provide 12 month series and breakdown by flow	Integ	EP	P2	+	Y	TA	Pass
TC-37	FR-09	Admin creates updates and deletes a place	Integ	State transition	P2	+	Y	TA	Pass
TC-38	FR-09	Non admin cannot create places events or stories	Integ	EP	P2	−	Y	TA	Pass
TC-39	FR-09	Admin creates and deletes an event and a story	Integ	EP	P3	+	Y	TA	Pass
TC-40	FR-07	Places filter by category and division	Integ	EP	P1	+	Y	TA	Pass
TC-41	FR-07	Search is case insensitive	Integ	EP	P2	+	Y	TA	Pass
TC-42	FR-07	Search with no match returns an empty list not an error	Integ	EP	P2	−	Y	TA	Pass
TC-43	FR-07	List limit is capped at 200	Integ	BVA	P2	+	Y	TA	Pass
TC-44	FR-07	Upcoming events end today or later	Integ	EP	P3	+	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-45	FR-08	Place detail has the fields listed in the srs	Integ	EP	P1	+	Y	TA	Pass
TC-46	FR-08	Unknown place returns 404	Integ	EP	P2	–	Y	TA	Pass
TC-47	FR-10	Categories report the number of places	Integ	EP	P2	+	Y	TA	Pass
TC-48	FR-11	Every place has coordinates inside bangladesh for the map	Integ	EP	P1	+	Y	TA	Pass
TC-49	FR-21	Products filter by category and price range	Integ	EP	P1	+	Y	TA	Pass
TC-50	FR-21	Products sort by price ascending and descending	Integ	EP	P2	+	Y	TA	Pass
TC-51	FR-21	Preorder filter returns only eligible in stock products	Integ	EP	P2	+	Y	TA	Pass
TC-52	FR-14	Guide updates profile languages and price	Integ	EP	P2	+	Y	TA	Pass
TC-53	FR-15	Public sees no coordinates unless available and sharing	Integ	Decision table	P1	–	Y	TA	Pass
TC-54	FR-15	Shared position is approximate for the public	Integ	EP	P2	+	Y	TA	Pass
TC-55	FR-16	Booking goes pending accepted finished completed and notifies	Integ	State transition	P1	+	Y	TA	Pass
TC-56	FR-16	Guide cannot skip from pending to completed	Integ	State transition	P1	–	Y	TA	Pass
TC-57	FR-16	Booking dates in the past or reversed are rejected	Integ	BVA	P1	–	Y	TA	Pass
TC-58	FR-16	Guide cannot be double booked for overlapping dates	Integ	EP	P2	–	Y	TA	Pass
TC-59	FR-17	Guide dashboard lists only the guides own bookings	Integ	EP	P2	+	Y	TA	Pass
TC-60	FR-18	Tourist creates a named trip with destinations	Integ	EP	P1	+	Y	TA	Pass
TC-61	FR-18	Trip needs a name and end not before start	Integ	BVA	P2	–	Y	TA	Pass
TC-62	FR-19	Days and stops keep their order and trip can be deleted	Integ	State transition	P1	+	Y	TA	Pass
TC-63	FR-19	Another tourist cannot view or delete my trip	Integ	EP	P1	–	Y	TA	Pass
TC-64	FR-20	Vendor adds edits and deletes a product with tags and specs	Integ	State transition	P1	+	Y	TA	Pass
TC-65	FR-20	Price must be above zero and stock not negative	Integ	BVA	P1	–	Y	TA	Pass
TC-66	FR-22	Cart add merges update remove and clear	Integ	State transition	P1	+	Y	TA	Pass
TC-67	FR-22	Unknown product cannot be added to cart	Integ	EP	P2	–	Y	TA	Pass
TC-68	FR-23	Checkout creates order clears cart and holds the advance	Integ	Use case	P1	+	Y	TA	Pass
TC-69	FR-23	Checkout with empty cart or missing address is rejected	Integ	EP	P1	–	Y	TA	Pass
TC-70	FR-24	Checkout fails when quantity exceeds stock and stock is unchanged	Integ	BVA	P1	–	Y	TA	Pass
TC-71	FR-24	Buying exactly the remaining stock succeeds and decrements to zero	Integ	BVA	P1	+	Y	TA	Pass
TC-72	FR-36	Delivery order moves through the srs states to completed	Integ	State transition	P1	+	Y	TA	Pass
TC-73	FR-36	Vendor cannot skip states or release the advance	Integ	State transition	P1	–	Y	TA	Pass
TC-74	FR-36	Order has a tracking number and estimated delivery	Integ	EP	P2	+	Y	TA	Fail
TC-75	FR-38	Order history is role aware	Integ	EP	P1	+	Y	TA	Pass
TC-76	FR-25	Preorder eligible product requires preparation time and pickup location	Integ	EP	P1	–	Y	TA	Pass
TC-77	FR-25	Tourist cannot preorder a product that is not preorder enabled	Integ	EP	P2	–	Y	TA	Pass
TC-78	FR-26	Preorder products are listed for a destination area	Integ	EP	P1	+	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-79	FR-27	Quantity must be a whole number within stock	Integ	BVA	P1	−	Y	TA	Pass (4/4)
TC-80	FR-28	Preorder stores 30 percent advance and 70 percent remaining	Integ	EP	P1	+	Y	TA	Pass
TC-81	FR-29	Advance is recorded with method payer number and trxid	Integ	EP	P1	+	Y	TA	Pass
TC-82	FR-29	Wallet advance without valid number or trxid is rejected	Integ	EP	P1	−	Y	TA	Pass (5/5)
TC-83	FR-29	Trxid of 6 and 20 characters and card without proof are accepted	Integ	BVA	P2	+	Y	TA	Pass
TC-84	FR-30	Each preorder gets a unique order number and qr code	Integ	EP	P1	+	Y	TA	Pass
TC-85	FR-31	Vendor moves preorder to ready and tourist is notified	Integ	State transition	P1	+	Y	TA	Pass
TC-86	FR-31	Vendor cannot skip from accepted to ready	Integ	State transition	P1	−	Y	TA	Pass
TC-87	FR-32	Wrong qr code is rejected	Integ	EP	P1	−	Y	TA	Pass
TC-88	FR-32	Qr code can be used only once	Integ	State transition	P1	−	Y	TA	Pass
TC-89	FR-33	Remaining payment by bkash is recorded and tourist confirms receipt	Integ	State transition	P1	+	Y	TA	Pass
TC-90	FR-33	Wallet pickup payment needs payer number and trxid	Integ	EP	P1	−	Y	TA	Pass
TC-91	FR-34	Cancellation before preparation refunds advance and restores stock	Integ	State transition	P1	+	Y	TA	Pass
TC-92	FR-34	Tourist cannot cancel once preparation has begun	Integ	State transition	P1	−	Y	TA	Pass
TC-93	FR-34	Refund amount follows the configured refund percentage	Integ	EP	P1	+	Y	TA	Pass
TC-94	FR-34	Cancellation needs a reason	Integ	EP	P2	−	Y	TA	Pass
TC-95	FR-35	Admin updates preorder settings and they are served publicly	Integ	EP	P1	+	Y	TA	Pass
TC-96	FR-35	Refund percentage outside 0 to 100 is rejected	Integ	BVA	P2	−	Y	TA	Pass
TC-97	FR-37	Full timeline reaches completed through every srs state	Integ	State transition	P1	+	Y	TA	Pass
TC-98	FR-39	Buyer reviews a purchased product	Integ	EP	P1	+	Y	TA	Pass
TC-99	FR-39	Rating must be between 1 and 5	Integ	BVA	P1	−	Y	TA	Pass (4/4)
TC-100	FR-39	Second review of the same item and review without purchase are refused	Integ	EP	P2	−	Y	TA	Pass
TC-101	FR-39	Tourist can review a heritage place	Integ	EP	P2	+	Y	TA	Fail
TC-102	FR-40	Average rating and count are recalculated	Integ	EP	P1	+	Y	TA	Pass
TC-103	FR-41	Reviews are listed with user date rating and comment	Integ	EP	P2	+	Y	TA	Pass
TC-104	FR-42	Notifications are created and can be marked read	Integ	State transition	P1	+	Y	TA	Pass
TC-105	FR-42	User cannot mark someone elses notification	Integ	EP	P2	−	Y	TA	Pass
TC-106	FR-43	Chat message raises unread count until read	Integ	Use case	P1	+	Y	TA	Pass
TC-107	FR-43	Empty message is rejected	Integ	EP	P2	−	Y	TA	Pass
TC-108	FR-16	Tourist reports cash and guide confirms it	Integ	State transition	P1	+	Y	TA	Pass
TC-109	FR-16	Guide records bkash payment directly and earnings show it received	Integ	State transition	P1	+	Y	TA	Pass
TC-110	FR-16	Guide payment rules are enforced	Integ	EP	P1	−	Y	TA	Pass
TC-111	FR-45	Admin ledger lists advances with payer number and trxid	Integ	EP	P1	+	Y	TA	Pass
TC-112	FR-50	Failed logins are logged with identifier and time	Integ	EP	P2	+	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-113	FR-50	Important user and payment events are kept in an audit log	Integ	EP	P2	+	Y	TA	Fail
TC-114	FR-03	Sql injection in search and login is harmless	Integ	Error guessing	P1	−	Y	TA	Pass
TC-115	FR-49	Errors never expose sql paths or stack traces	Integ	Error guessing	P1	−	Y	TA	Pass
TC-116	FR-02	Passwords are stored only as bcrypt hashes	Integ	EP	P1	+	Y	TA	Pass
TC-117	FR-51	Schema and data can be restored from the committed sql	Integ	EP	P2	+	Y	TA	Pass
TC-118	FR-20	Vendor uploads a product image that is served back	Integ	EP	P1	+	Y	TA	Pass
TC-119	FR-20	Non image disguised as png and anonymous uploads are rejected	Integ	Error guessing	P1	−	Y	TA	Pass
TC-120	FR-01	Vendor uploads a verification document that only admins can list	Integ	EP	P2	+	Y	TA	Pass
TC-121	FR-06	Vendor completes shop profile	Integ	EP	P2	+	Y	TA	Pass
TC-122	FR-06	Incomplete shop profile is rejected	Integ	EP	P2	−	Y	TA	Pass
TC-123	FR-07	Local foods for a district list that district first with bangla names	Integ	EP	P2	+	Y	TA	Pass
TC-124	FR-09	Admin adds edits and removes a local food	Integ	State transition	P2	+	Y	TA	Pass
TC-125	FR-09	Admin sets homepage hero image and rejects invalid urls	Integ	EP	P3	±	Y	TA	Pass
TC-126	FR-34	Tourist cannot choose their own refund amount	Integ	Error guessing	P1	−	Y	TA	Pass
TC-U01	FR-28	Srs example 1000 taka splits into 300 advance and 700 balance	Unit	EP	P1	+	Y	TA	Pass
TC-U02	FR-28	Each flow uses its own terms	Unit	Decision table	P1	+	Y	TA	Pass (3/3)
TC-U03	FR-28	Zero amount gives all zero shares	Unit	BVA	P2	+	Y	TA	Pass
TC-U04	FR-28	Advance equal to commission releases nothing early	Unit	BVA	P2	+	Y	TA	Pass
TC-U05	FR-28	Full advance leaves no cash balance	Unit	BVA	P2	+	Y	TA	Pass
TC-U06	FR-28	Odd poisha amount is rounded and shares still add up	Unit	BVA	P2	+	Y	TA	Pass
TC-U07	FR-28	Rejects invalid amounts or percentages	Unit	EP	P1	−	Y	TA	Pass (4/4)
TC-U08	FR-29	Valid bangladeshi mobile numbers are normalised	Unit	EP	P1	+	Y	TA	Pass (5/5)
TC-U09	FR-29	Invalid numbers are rejected	Unit	BVA	P1	−	Y	TA	Pass (6/6)
TC-U10	FR-26	Division name resolves to whole division	Unit	EP	P1	+	Y	TA	Pass
TC-U11	FR-26	Old spelling resolves to current district	Unit	EP	P1	+	Y	TA	Pass
TC-U12	FR-26	Landmark inside destination name resolves to its district	Unit	EP	P2	+	Y	TA	Pass
TC-U13	FR-26	Prefer district keeps same named district	Unit	EP	P2	+	Y	TA	Pass
TC-U14	FR-21	Unknown or blank area resolves to null	Unit	EP	P2	−	Y	TA	Pass
TC-U15	FR-21	District query is widened to its division unless exact	Unit	EP	P3	+	Y	TA	Pass
TC-U16	FR-20	Own upload paths and https urls are accepted as images	Unit	EP	P2	+	Y	TA	Pass
TC-U17	FR-20	Scripts other schemes and malformed paths are not images	Unit	EP	P1	−	Y	TA	Pass
TC-U18	FR-20	Image reference longer than 500 chars is rejected	Unit	BVA	P3	−	Y	TA	Pass
TC-U19	FR-15	Json booleans become 0 or 1 at any depth	Unit	EP	P2	+	Y	TA	Pass
TC-U20	FR-02	Only a sha256 hash of the token is stored	Unit	EP	P1	+	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-U21	FR-30	Generated ids are prefixed and unique	Unit	EP	P2	+	Y	TA	Pass
TC-F01	FR-29	Accepts a complete bKash proof, including +880 numbers	Unit	EP	P1	+	Y	TA	Pass
TC-F02	FR-29	Asks for the number, then the TrxID, naming the wallet	Unit	EP	P1	–	Y	TA	Pass
TC-F03	FR-29	Phone	Unit	BVA	P1	–	Y	TA	Pass (5/5)
TC-F04	FR-29	TrxID	Unit	BVA	P1	–	Y	TA	Pass (5/5)
TC-F05	FR-33	Card and cash need no proof	Unit	EP	P2	+	Y	TA	Pass
TC-F06	FR-29	Shows number and TrxID inputs with the amount only for wallets	Unit	EP	P2	+	Y	TA	Pass
TC-F07	FR-33	Vendor view asks for the customer\	Unit	EP	P3	+	Y	TA	Pass
TC-F08	FR-23	Labels every payment method	Unit	EP	P3	+	Y	TA	Pass
TC-F09	FR-37	Pre-order status	Unit	EP	P2	+	Y	TA	Pass (12/12)
TC-F10	FR-36	Delivery status	Unit	EP	P2	+	Y	TA	Pass (8/8)
TC-F11	FR-37	Unknown or empty status degrades gracefully	Unit	EP	P2	–	Y	TA	Pass
TC-F12	FR-16	Terminal states use success / danger colours	Unit	EP	P3	+	Y	TA	Pass
TC-F13	FR-37	Badge renders the label and a custom label override	Unit	EP	P3	+	Y	TA	Pass
E2E-01	FR-11, FR-12, FR-13	Map shows heritage places, layer toggles and marker card links to the place	E2E	Use case	P1	+	Y	TA	Pass (2/2)
E2E-02	FR-27, FR-28, FR-29, FR-30	Tourist pre-orders with bKash proof and gets an order number	E2E	Use case	P1	+	Y	TA	Pass (2/2)
E2E-03	FR-45	Admin sees advances with payer number and TrxID on Payments & Advances	E2E	Use case	P1	+	Y	TA	Pass (2/2)

5.2 Detailed test cases (priority-1)

All 96 priority-1 cases are automated; the blocks below specify them in the template format. The expected result is taken from the SRS (or from the payment terms the SRS refers to), not from the code.

Test case ID / title	TC-03 — password shorter than 8 characters is rejected
Requirement / acceptance criterion	FR-01 / §5.5: passwords of at least 8 characters
Level / technique / priority	Integration / BVA / P1
Preconditions	No account with the e-mail exists
Test data	passwords "Abc1234" (7), "Abc12345" (8), "Abc123456" (9), ""
Steps	POST /api/auth/register for each password
Expected result (from SRS)	7 chars and empty → 422; 8 and 9 chars → 201
Actual result	As expected
Status	Pass
Automated test location	tests/integration/AuthTest.php › test_TC_03_password_shorter_than_8_characters_is_rejected
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-09 — sixth failed login within 15 minutes is throttled
Requirement / acceptance criterion	FR-02: account protection against guessing
Level / technique / priority	Integration / State transition / P1

Preconditions	Fresh unregistered e-mail
Test data	6 logins with password "wrong"
Steps	POST /api/auth/login six times
Expected result (from SRS)	Codes 401 ×5, then 429 (locked for 15 min)
Actual result	As expected
Status	Pass
Automated test location	tests/integration/AuthTest.php › test_TC_09_sixth_failed_login_within_15_minutes_is_throttled
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-13 — valid otp resets password and signs out old sessions
Requirement / acceptance criterion	FR-05: 6-digit OTP valid 15 minutes resets the password
Level / technique / priority	Integration / EP / P1
Preconditions	Registered tourist; OTP hash inserted, expiring in 15 min
Test data	otp 482913, new password NewSecret99
Steps	POST /api/auth/reset-password; log in with new password; call /auth/me with old token
Expected result (from SRS)	200; login with new password 200; old token 401
Actual result	As expected
Status	Pass
Automated test location	tests/integration/AuthTest.php › test_TC_13_valid_otp_resets_password_and_signs_out_old_sessions
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-22 — tourist cannot read another tourists preorder
Requirement / acceptance criterion	FR-49: 403 for requests outside the user's ownership
Level / technique / priority	Integration / Error guessing / P1
Preconditions	Tourist A owns a pre-order; tourist B registered
Test data	A's pre-order ID
Steps	B: GET and PUT /api/preorders/{id}
Expected result (from SRS)	403 for both
Actual result	As expected
Status	Pass
Automated test location	tests/integration/AccessControlTest.php › test_TC_22_tourist_cannot_read_another_tourists_preorder
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-29 — rejecting a pending vendor suspends the account
Requirement / acceptance criterion	FR-04: rejection suspends the account
Level / technique / priority	Integration / State transition / P1

Preconditions	Pending vendor
Test data	status = rejected
Steps	Admin rejects; vendor logs in
Expected result (from SRS)	Login 403 (account suspended)
Actual result	As expected
Status	Pass
Automated test location	tests/integration/AdminTest.php › test_TC_29_rejecting_a_pending_vendor_suspends_the_account
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-53 — public sees no coordinates unless available and sharing
Requirement / acceptance criterion	FR-15: coordinates exposed only when available AND sharing
Level / technique / priority	Integration / Decision table / P1
Preconditions	Seed guide
Test data	availability / sharing combinations
Steps	Toggle flags; GET /api/guides/{id} as public
Expected result (from SRS)	null, null, then approximate coordinates
Actual result	As expected
Status	Pass
Automated test location	tests/integration/GuideTripTest.php › test_TC_53_public_sees_no_coordinates_unless_available_and_sharing
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-68 — checkout creates order clears cart and holds the advance
Requirement / acceptance criterion	FR-23: order created from the cart with address and method; 20% advance (payment terms)
Level / technique / priority	Integration / Use case / P1
Preconditions	Verified vendor product ৳400; empty cart
Test data	qty 2, card
Steps	Add to cart; POST /api/orders
Expected result (from SRS)	201, status placed, total ৳920 (incl. ৳120 delivery), advance ৳184 held, cart empty
Actual result	As expected
Status	Pass
Automated test location	tests/integration/MarketplaceTest.php › test_TC_68_checkout_creates_order_clears_cart_and_holds_the_advance
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-70 — checkout fails when quantity exceeds stock and stock is unchanged
Requirement / acceptance criterion	FR-24 / §5.5: checkout fails if stock is insufficient

Level / technique / priority	Integration / BVA / P1
Preconditions	Product with stock 3
Test data	qty 4 (stock + 1)
Steps	Checkout
Expected result (from SRS)	409/422 and stock still 3
Actual result	As expected
Status	Pass
Automated test location	tests/integration/MarketplaceTest.php › test_TC_70_checkout_fails_when_quantity_exceeds_stock_and_stock_is_unchanged
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-80 — preorder stores 30 percent advance and 70 percent remaining
Requirement / acceptance criterion	FR-28 / §4.2.1: ৳1,000 → advance ৳300, remaining ৳700
Level / technique / priority	Integration / EP / P1
Preconditions	Pre-order product ৳1,000
Test data	qty 1
Steps	POST /api/preorders
Expected result (from SRS)	total 1000, advancePaid 300, remainingAmount 700, advancePercent 30
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PreorderTest.php › test_TC_80_preorder_stores_30_percent_advance_and_70_percent_remaining
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-82 — wallet advance without valid number or trxid is rejected
Requirement / acceptance criterion	FR-29: wallet advance needs payer number and TrxID
Level / technique / priority	Integration / EP / P1
Preconditions	Pre-order product
Test data	missing/invalid number; TrxID of 5 and 21 chars
Steps	POST /api/preorders per case
Expected result (from SRS)	422 for each
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PreorderTest.php › test_TC_82_wallet_advance_without_valid_number_or_trxid_is_rejected
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-88 — qr code can be used only once
Requirement / acceptance criterion	FR-32 / §5.3: QR code usable once

Level / technique / priority	Integration / State transition / P1
Preconditions	Pre-order in qr_verified
Test data	the same QR code
Steps	Vendor verifies the QR a second time
Expected result (from SRS)	409/422
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PreorderTest.php › test_TC_88_qr_code_can_be_used_only_once
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-92 — tourist cannot cancel once preparation has begun
Requirement / acceptance criterion	FR-34: cancellation only before preparation begins
Level / technique / priority	Integration / State transition / P1
Preconditions	Pre-order in preparing
Test data	reason "Too late"
Steps	Tourist PUT status cancelled
Expected result (from SRS)	Refused (403/409/422)
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PreorderTest.php › test_TC_92_tourist_cannot_cancel_once_preparation_has_begun
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-93 — refund amount follows the configured refund percentage
Requirement / acceptance criterion	FR-34: configured refund policy applied (seed 80%)
Level / technique / priority	Integration / EP / P1
Preconditions	Pre-order ৳1,000 in vendor_accepted
Test data	reason "Change of plans"
Steps	Tourist cancels
Expected result (from SRS)	refundAmount ৳240 (80% of ৳300)
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PreorderTest.php › test_TC_93_refund_amount_follows_the_configured_refund_percentage
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-126 — tourist cannot choose their own refund amount
Requirement / acceptance criterion	§5.3: no unauthorised modification of payment records
Level / technique / priority	Integration / Error guessing / P1

Preconditions	Pre-order in vendor_accepted
Test data	refundAmount 99999 in the request
Steps	Tourist cancels with refundAmount
Expected result (from SRS)	200 and refundAmount ৳240 (client value ignored)
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PreorderTest.php › test_TC_126_tourist_cannot_choose_their_own_refund_amount
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	TC-114 — sql injection in search and login is harmless
Requirement / acceptance criterion	§5.3: prepared statements prevent SQL injection
Level / technique / priority	Integration / Error guessing / P1
Preconditions	Seeded places
Test data	search "' OR 1=1 --", login e-mail with quote
Steps	GET /api/places?search=…; POST /api/auth/login
Expected result (from SRS)	Search returns fewer than all places; login 401/422
Actual result	As expected
Status	Pass
Automated test location	tests/integration/PaymentsSecurityTest.php › test_TC_114_sql_injection_in_search_and_login_is_harmless
Executed by / date	npm test / npm run test:e2e, 26 October 2026

Test case ID / title	E2E-02 — tourist pre-orders with bKash proof and gets an order number
Requirement / acceptance criterion	FR-27 – FR-30: pre-order checkout with advance payment and order number
Level / technique / priority	System/E2E / Use case / P1
Preconditions	Tourist signed in; product po-5
Test data	bKash 01712345678, TrxID E2E7XK2L5M
Steps	Pickup info → summary → pay without proof (blocked) → enter proof → pay
Expected result (from SRS)	Validation message, then confirmation with an order number HByymmddXXXXX
Actual result	As expected
Status	Pass
Automated test location	tests/e2e/journeys.spec.ts › E2E-02 tourist pre-orders with bKash proof and gets an order number
Executed by / date	npm test / npm run test:e2e, 26 October 2026

The remaining 80 priority-1 cases follow the same structure; their preconditions, data and expected values are in the test code named in §5.1.

5.3 Equivalence partitions and boundary values

Input	Valid partition(s)	Invalid partition(s)	Boundary values tested	TC IDs

Input	Valid partition(s)	Invalid partition(s)	Boundary values tested	TC IDs
Password length (register)	≥ 8 characters	< 8, empty	7, 8, 9, empty	TC-03
Reset OTP	exactly 6 digits, unexpired	5 / 7 digits, letters, expired	5, 6, 7 digits; expiry −1 s	TC-14, TC-15
BD mobile number (payer)	01[3-9] + 8 digits, ±880 prefix	operator 0–2, 10 / 12 digits, letters, empty	0171234567 (10), 01712345678 (11), 017123456789 (12), 012…, 013…, 019…	TC-U08, TC-U09, TC-F03
Wallet TrxID	6–20 letters/digits	< 6, > 20, symbols, empty	5, 6, 20, 21 chars	TC-82, TC-83, TC-F04
Review rating	1 – 5	< 1, > 5	0, 1, 5, 6	TC-99
Product price / stock	price > 0; stock ≥ 0 integer	price 0, stock −1, unknown category	0 / 0.01 price; −1 / 0 stock	TC-65
Order / pre-order quantity	1 – stock	0, negative, fraction, > stock	stock, stock + 1	TC-70, TC-71, TC-79
Refund percentage (settings)	0 – 100	< 0, > 100	−1, 101	TC-96
Booking / trip dates	start ≥ today, end ≥ start	past start, end < start	yesterday; end = start − 1	TC-57, TC-61
List limit	1 – 200	> 200	1, 100000	TC-43
Image reference length	≤ 500 chars	> 500	500, 501	TC-U18
Money split (advance, commission)	0 ≤ commission ≤ advance ≤ 100	commission > advance, advance > 100, negative	advance = commission; 100%; ৳0	TC-U03 – TC-U07

5.4 Decision table / state transition

Decision table — guide position visible to the public (FR-15)

Rule	Available	Location sharing	Verified	Public sees	TC
R1	Y	N	Y	No position	TC-53
R2	N	Y	Y	No position	TC-53
R3	Y	Y	Y	Approximate (~1 km)	TC-53, TC-54
R4	any	any	self / admin	Exact	TC-54 (owner view)

State transition — pre-order (FR-31 – FR-34, FR-37)

From	Event / actor	To	Valid?	TC
advance_paid	vendor accepts	vendor_accepted	Valid	TC-85
vendor_accepted	vendor skips to ready	—	Invalid (422)	TC-86
vendor_accepted	tourist moves forward	—	Invalid (403)	TC-27
vendor_accepted	tourist cancels	cancelled, refund 80%	Valid	TC-91, TC-93
preparing	tourist cancels	—	Invalid (409)	TC-92
arrived	wrong QR	—	Invalid (422)	TC-87
qr_verified	same QR again	—	Invalid	TC-88
pickup_payment_pending	vendor records bKash + TrxID	picked_up	Valid	TC-89
pickup_payment_pending	wallet without TrxID	—	Invalid (422)	TC-90
picked_up	tourist confirms receipt	completed, advance released	Valid	TC-89, TC-97

State transition — delivery order (FR-36) and guide booking (FR-16)

Order: placed → confirmed → preparing → shipped → out_for_delivery → delivered → completed (tourist only) — TC-72; skipping states and vendor release of the advance are refused — TC-73. Booking: pending → accepted → awaiting_confirmation → completed — TC-55; pending → completed refused — TC-56; overlapping accepted bookings refused — TC-58. Login lock-out after 5 failures — TC-09.

6. Test Execution Results

6.1 Summary

Executed on v1.0.1-test (1d45c59). Counts are executions (data-provider rows and browser projects count separately).

Level / type	Total	Passed	Failed	Blocked	Skipped	Pass %
Unit	74	74	0	0	0	100%
Integration / API	159	156	3	0	0	98.1%
System / E2E	6	6	0	0	0	100%
Acceptance	covered by the FR-tagged integration and E2E cases above					
Total	239	236	3	0	0	98.7%

6.2 Continuous-integration evidence

Workflow file	Not added in v1.0 (team decision; planned for Experiment 12)
CI run URL (tagged commit)	—
Commit SHA of run	1d45c59 (local runs; fresh-clone run of the tag in A9)
Run status	Local: Passed except the 3 open-defect tests (§6.3)
Branch protection requires checks	No
Consecutive green runs (flakiness check)	E2E: 3 consecutive green runs (A8)

N/A justification: the guideline makes GitHub Actions a Professional-tier item; this report targets Baseline, which requires that all unit and integration tests run with one documented command (met, verified on a fresh clone).

6.3 Analysis of failed, blocked or skipped tests

TC ID	Status	Reason	Linked defect	Action / plan
TC-74	Fail	trackingNumber null; estimatedDelivery null until the vendor sets it	DEF-08	Deferred (P3); test kept failing as the acceptance check
TC-113	Fail	Only failed logins are recorded (auth_attempts)	DEF-10	Deferred (P3); test kept failing as the acceptance check
TC-101	Fail	422 'Place reviews require a verified visit record, which is not supported yet.'	DEF-09	Deferred (P2); test kept failing as the acceptance check

7. Code Coverage

API coverage was measured with Xdebug in the PHP server process during the integration run (783 requests) and merged with the unit-test coverage. Branch coverage is not measured (line mode).

Module / layer	Line %	Branch %	Function %	Target	Met?

Module / layer	Line %	Branch %	Function %	Target	Met?
Data access / config	76.2% (48/63)	—	100%	—	—
Controllers / API	78.4% (2273/2900)	—	— (procedural)	—	—
Business logic / helpers	92.7% (228/246)	—	— (procedural)	≥ 60%	Met
Router (index.php)	97.1% (134/138)	—	— (procedural)	—	—
UI components (unit-tested modules)	91.66%	97.36%	80%	—	—
Overall (API)	80.2% (2683/3347)	—	—	≥ 60%	Met

Uncovered code — justification

File	Line %	Why not covered
controllers/trips_controller.php	50.3%	Guide-shared trip plans (trip ↔ guide response, guide fee) are tested only through bookings; residual risk §11.2
controllers/uploads_controller.php	50.6%	Profile images and document download error paths
controllers/admin_controller.php	56.4%	Pending-list endpoints and document review screens; covered by the API smoke test, not by tagged cases
src/pages/** (React pages)	—	Not unit-tested; exercised by the E2E journeys only

Mutation testing (Distinction tier)

Tool / module	Not performed (Baseline target)
Mutants generated / killed / survived	—
Mutation score	—
Survivors analysed (examples)	Manual planted-bug check: reverting each fix in §9 made its test fail (A11), showing the suite detects those faults

8. Non-Functional Testing

8.1 Performance

Scenario	Load profile	Target (NFR)	p95 measured	Error rate	Result
PT-01 GET /api/places	5 concurrent, 60 requests	< 2000 ms (NFR-01)	862 ms (p50 679)	0/60	Pass
PT-02 GET /api/products?limit=50	5 concurrent, 60 requests	< 2000 ms (NFR-01)	1103 ms (p50 981)	0/60	Pass
PT-03 GET /api/guides	5 concurrent, 60 requests	< 2000 ms (NFR-01)	838 ms (p50 650)	0/60	Pass
PT-04 GET /api/places/p1	5 concurrent, 60 requests	< 2000 ms (NFR-01)	473 ms (p50 334)	0/60	Pass
PT-05 GET /api/events?upcoming=1	5 concurrent, 60 requests	< 2000 ms (NFR-01)	389 ms (p50 298)	0/60	Pass

Tool and script: tests/performance/p95.mjs against https://heritagebd-flax.vercel.app (Vercel → Render free plan → Supabase), 2026-10-26 07:28 UTC, warm-up 490 ms excluded. The SRS gives no number for "fast response"; 2 s is the team's target and should be added to the SRS. A first run during a Render deploy returned 502 for every request (DEF-12).

8.2 Security testing

Check	Tool	Findings C / H / M / L	Action taken	Status
Dependency vulnerability audit (npm)	npm audit (npm 11.19)	0 / 0 / 0 / 0	Vitest 3.x advisories (2 critical in dev tooling) fixed by upgrading to 4.1.12	Pass

Check	Tool	Findings C / H / M / L	Action taken	Status
Dependency audit (Composer)	composer audit 2.8	0 / 0 / 0 / 0	—	Pass
Secret scanning	gitleaks 8.30.1 (history + working tree)	0 leaks	—	Pass
Transport security	curl (SEC-01)	HTTP→HTTPS 308/301, HSTS present; no X-Frame-Options / nosniff / CSP	DEF-16 logged	Partial
CORS policy	curl (SEC-02)	Access-Control-Allow-Origin: *	DEF-15: set CORS_ORIGIN on Render	Fail
Dynamic scan (DAST)	OWASP ZAP baseline	Not run (Professional tier; planned Exp 10)	—	Not run

OWASP-based manual security test checklist

Risk area	Test performed	TC ID	Result
Broken access control	Tourist B reads/changes tourist A's pre-order, order, chat, trip; guide edits another guide; vendor edits another vendor's product; tourist forces vendor-only states	TC-20 – TC-27, TC-63, TC-105	Pass
Authentication failures	Lock-out after 5 failures, weak password rejected, expired token, logout revokes token, suspended account	TC-03, TC-09 – TC-12	Pass
Injection (SQL)	' OR 1=1 -- in search and login	TC-114	Pass
Cross-site scripting (XSS)	Script URLs rejected as images; React escapes text output by default (no dangerouslySetInnerHTML in pages)	TC-U17, TC-125	Pass
Cryptographic failures	HTTPS enforced; passwords bcrypt; only SHA-256 of tokens stored	SEC-01, TC-116, TC-U20	Pass
Security misconfiguration	No SQL/paths/stack traces in errors; PHP warnings not shown	TC-115, TC-16	Pass
Payment record tampering	Client-chosen refund amount, wallet payment without proof	TC-126, TC-110	Pass
Vulnerable components	See dependency audit above	—	Pass

8.3 Static analysis and quality gate

Experiment 10 deliverable. For v1.0 the Baseline linter check applies: TypeScript compiler (strict) and php -l on all 53 PHP files report 0 errors (A10). SonarCloud metrics will be added in v1.1.

Metric	Value	Gate threshold	Pass?
Linter errors (tsc, php -l)	0	0	Pass
Bugs / Vulnerabilities / Code smells (SonarCloud)	v1.1	—	—
Security hotspots reviewed	v1.1	100%	—
Duplication %	v1.1	≤ 3%	—
Quality gate status	v1.1	—	—

8.4 Accessibility and compatibility

Screen / page	Lighthouse a11y	axe critical	Edge (desktop)	Firefox	Mobile viewport	Notes
LH-01 Landing (/)	83	not run	Loads, audited	Not tested	Not tested	perf 53, best practices 100
LH-02 Places (/places)	85	not run	Loads, audited	Not tested	Not tested	perf 78, best practices 100
LH-03 Place detail (/places/p7)	88	not run	Loads, audited	Not tested	Not tested	perf 76, best practices 96

Screen / page	Lighthouse a11y	axe critical	Edge (desktop)	Firefox	Mobile viewport	Notes
LH-04 Shop (/shop)	82	not run	Loads, audited	Not tested	Not tested	perf 75, best practices 100
LH-05 Guides (/guides)	85	not run	Loads, audited	Not tested	Not tested	perf 71, best practices 100
LH-06 Login (/login)	95	not run	Loads, audited	Not tested	Pass (E2E sign-in)	perf 96, best practices 100
Map (/map)	—	not run	Pass (E2E-01)	Not tested	Pass (E2E-01)	markers, layer toggle, marker card
Pre-order checkout	—	not run	Pass (E2E-02)	Not tested	Pass (E2E-02)	bKash proof validation
Admin › Payments & Advances	—	not run	Pass (E2E-03)	Not tested	Pass (E2E-03)	ledger filter

Lighthouse 12.8.2 on Edge 155 headless against the deployed site. Failing audits (DEF-11): colour contrast (74 elements), icon buttons without accessible names (24), links without discernible names (20), selects without labels (4), heading order (4), one small touch target on /login. E2E journeys passed on desktop Edge and on a Pixel 7 viewport; Firefox and Safari were not available (§11.2).

8.5 Usability

Participants (number, profile)	Not yet conducted — to be done with 3 students from other sections who have not seen the app
Tasks given	T1: register as a tourist; T2: pre-order a product and pay the advance by bKash; T3: find a guide on the map and send a booking request
Task completion rate	—
Average time per task	—
SUS score (optional)	—
Key issues found → Issue #	—

9. Defect Report

9.1 Summary by severity

Severity	Found	Fixed & verified	Open	Deferred (justified)
S1 — Critical	1	1	0	0
S2 — Major	6	5	1	1
S3 — Minor	8	3	5	2
S4 — Trivial	1	0	1	0
Total	16	9	7	3

Defects found by own tests	10
Defects found by cross-team testing (label found-by:)	0 (cross-team session not held; Professional tier)
Defects found by tools (scanners / generators)	4
Defects found by review / users	2

9.2 Defect log

Defects are tracked in docs/testing/defects.json (not yet mirrored as GitHub Issues). "Fix" is the commit; the regression test failed before the fix and passes after it (A11).

ID	Title	Sev	Pri	Found by (TC / level)	Status	Fix	Regression TC
DEF-07	Tourist can set any refund amount when cancelling a pre-order	S1	P1	TC-126 / Integration (error guessing)	Closed	99d71bf	TC-126
DEF-04	Tourist can cancel a pre-order after preparation has started	S2	P1	TC-92 / Integration (state transition)	Closed	99d71bf	TC-92
DEF-05	Refund amount not calculated from the configured refund percentage	S2	P1	TC-93 / Integration (EP)	Closed	99d71bf	TC-93
DEF-03	Rejected vendor account stays active	S2	P1	TC-29 / Integration (state transition)	Closed	99d71bf	TC-29
DEF-13	Reporting a guide payment fails on PostgreSQL	S2	P1	Code review while building the guide cash flow	Closed	8bf0f55	TC-108
DEF-14	Bengali names shown as garbled text (mojibake)	S2	P1	User report; data scan of the live API	Closed	data fix on Su…	TC-123
DEF-02	PHP warnings leak into API responses	S3	P2	TC-16 / Integration (EP)	Closed	99d71bf	TC-16
DEF-06	Pre-order settings accept out-of-range values	S3	P2	TC-96 / Integration (BVA)	Closed	99d71bf	TC-96
DEF-01	API smoke test out of date after required payment proof	S3	P2	backend/tests/api_smoke.php (existing suite)	Closed	99d71bf	api_smoke.php 'create pre-order'
DEF-09	Place reviews cannot be submitted	S2	P2	TC-101 / Integration (EP)	Open (deferred)	—	TC-101
DEF-08	No tracking number or automatic 5-day delivery estimate	S3	P3	TC-74 / Integration (EP)	Open (deferred)	—	TC-74
DEF-10	No audit log of user and payment events	S3	P3	TC-113 / Integration (EP)	Open (deferred)	—	TC-113
DEF-11	Accessibility: low contrast, unnamed icon buttons/links, unlabelled selects	S3	P2	Lighthouse 12.8.2 (tool)	Open	—	Lighthouse re-run
DEF-15	API accepts requests from any origin (CORS *)	S3	P2	SEC-02 / manual security check	Open	Set CORS_ORIGI…	SEC-02
DEF-16	No clickjacking / MIME-sniffing / CSP headers on the site	S4	P3	SEC-01 / manual security check	Open	—	SEC-01
DEF-12	API returns 502 for about a minute after each deploy	S3	P3	Performance run (p95.mjs)	Open	—	p95.mjs (errors = 0)

9.3 Detailed reports for S1 and S2 defects

Issue # / title	DEF-07 — Tourist can set any refund amount when cancelling a pre-order
Severity / priority	S1 / P1 — Closed
Environment (browser, OS, commit)	PHP 8.2.35 API on PGlite (PostgreSQL 17), Windows 11, base commit 72586ab
Steps to reproduce	1. Log in as a tourist with a pre-order in vendor_accepted 2. PUT /api/preorders/{id} {status: cancelled, cancellationReason: 'x', refundAmount: 99999} 3. GET the pre-order
Expected result (SRS ref.)	Refund computed by the server from the configured policy (SRS §5.3: prevent unauthorised modification of payment records; FR-34)

Actual result	refundAmount 99999 stored and shown as the refund
Evidence	regression-before-after.txt: 'Failed asserting that 99999 matches expected 240'
Root cause	preorders_update copied refundAmount from any participant's request body into the row (generic field map) without a role check.
Fix (PR / commit)	99d71bf
Regression test (fails before, passes after)	TC-126

Issue # / title	DEF-04 — Tourist can cancel a pre-order after preparation has started
Severity / priority	S2 / P1 — Closed
Environment (browser, OS, commit)	PHP 8.2.35 API on PGlite, Windows 11, base 72586ab
Steps to reproduce	1. Vendor moves a pre-order to preparing 2. Tourist sends status cancelled with a reason
Expected result (SRS ref.)	Refused: cancellation only before preparation begins (FR-34, SRS §4.2.6)
Actual result	200 OK, pre-order cancelled and advance marked refunded
Evidence	regression-before-after.txt: TC-92 'Failed asserting that an array contains 200'
Root cause	PREORDER_TRANSITIONS allowed cancelled from every non-final state for every participant; no tourist-specific rule.
Fix (PR / commit)	99d71bf
Regression test (fails before, passes after)	TC-92

Issue # / title	DEF-05 — Refund amount not calculated from the configured refund percentage
Severity / priority	S2 / P1 — Closed
Environment (browser, OS, commit)	PHP 8.2.35 API on PGlite, Windows 11, base 72586ab
Steps to reproduce	1. Pre-order of ৳1,000 (advance ৳300), policy 80% 2. Tourist cancels before preparation
Expected result (SRS ref.)	refundAmount = 80% × ৳300 = ৳240 (FR-34, SRS §4.2.6)
Actual result	refundAmount null
Evidence	regression-before-after.txt: 'Failed asserting that null matches expected 240'
Root cause	Known limitation (SRS §6.5): settings were stored but never applied on cancellation.
Fix (PR / commit)	99d71bf
Regression test (fails before, passes after)	TC-93

Issue # / title	DEF-03 — Rejected vendor account stays active
Severity / priority	S2 / P1 — Closed
Environment (browser, OS, commit)	PHP 8.2.35 API on PGlite, Windows 11, base 72586ab
Steps to reproduce	1. Register a vendor 2. Admin PUT /api/admin/vendors/{id}/verify {status: rejected} 3. Vendor logs in

Expected result (SRS ref.)	Login refused: rejection suspends the account (FR-04, SRS §4.5)
Actual result	200 OK, vendor logged in
Evidence	regression-before-after.txt: TC-29 'Failed asserting that 200 is identical to 403'
Root cause	admin_vendor_verify always set status = 'active'; the guide version set 'suspended' on rejection.
Fix (PR / commit)	99d71bf
Regression test (fails before, passes after)	TC-29

Issue # / title	DEF-13 — Reporting a guide payment fails on PostgreSQL
Severity / priority	S2 / P1 — Closed
Environment (browser, OS, commit)	Production: Render API + Supabase PostgreSQL
Steps to reproduce	1. Tourist reports a cash payment for a finished tour (POST /api/guide-payments)
Expected result (SRS ref.)	201 Created, payment reported (FR-16)
Actual result	500: INSERT used the double-quoted "reported", which PostgreSQL reads as a column name
Evidence	TC-108 passes on PostgreSQL
Root cause	MySQL-only SQL (double-quoted string literal) missed in the PostgreSQL migration.
Fix (PR / commit)	8bf0f55
Regression test (fails before, passes after)	TC-108

Issue # / title	DEF-14 — Bengali names shown as garbled text (mojibake)
Severity / priority	S2 / P1 — Closed
Environment (browser, OS, commit)	Production (Supabase data), all browsers
Steps to reproduce	1. Open a place or local food page
Expected result (SRS ref.)	Bangla names readable (SRS §5.4: Bangla-capable storage)
Actual result	'αª¼αªù…' instead of 'বগুড়ার দই'
Evidence	Scan of 79 live items: 0 garbled after fix; TC-123 asserts Bengali script
Root cause	The XAMPP dump was written through the Windows console (code page 437).
Fix (PR / commit)	data fix on Supabase (79 values) + 8bf0f55 (fonts, dump command)
Regression test (fails before, passes after)	TC-123

Issue # / title	DEF-09 — Place reviews cannot be submitted
Severity / priority	S2 / P2 — Open (deferred)
Environment (browser, OS, commit)	PHP 8.2.35 API on PGlite; also production
Steps to reproduce	1. Tourist POST /api/reviews {targetType: place, targetId: p1, rating: 5, content}
Expected result (SRS ref.)	201 (FR-39: reviews for places, products and guides)

Actual result	422 'Place reviews require a verified visit record, which is not supported yet.'
Evidence	junit-php.xml TC-101
Root cause	Design choice to require a visit record that does not exist yet.
Fix (PR / commit)	Not fixed (deferred)
Regression test (fails before, passes after)	TC-101

10. Tool-Assisted Testing Log

Disclosure: the test suite, harness and this report were produced with Claude Code (Anthropic), an AI coding assistant, working in the team's repository at the team's request. Every test case is therefore marked TA in §5.1. Expected values were written from the SRS and the configured payment terms before the tests were run, and every failing test was investigated before its expected value was accepted. Team members must review each test before the viva (Guideline §7).

#	Tool	Asked / generated	Own version first?	How the expected value was verified	Decision	Reason
1	Claude Code	Integration tests for all 51 FRs (tests/integration)	N — generated from the SRS	Compared with SRS §4–§5 acceptance criteria	Accepted	Expected values trace to the SRS text quoted in each test
2	Claude Code	TC-36 admin report series	N	SRS FR-47 gives no period	Edited	Tool assumed a 12-month admin chart; not in the SRS, so the assertion was removed
3	Claude Code	TC-37 place update	N	API contract (visitInfo)	Edited	Tool sent entryFee at the top level; the API nests it in visitInfo
4	Claude Code	TC-65 invalid category	N	Re-read request data	Edited	PHP array union kept the valid category, so the negative case tested nothing
5	Claude Code	TC-23, TC-59, TC-109, TC-110	N	Ran classes alone and in full	Edited	Tests depended on seed data or on other tests; now create their own data
6	Claude Code	E2E-01 map journey	N	3 consecutive runs	Edited	Counted markers before guide data loaded (flaky)
7	Claude Code	TC-74, TC-101, TC-113	N	SRS §4.4, FR-39, FR-50	Accepted (failing)	App differs from the SRS; kept failing as acceptance checks (DEF-08 – DEF-10)

Reflection

The clearest wrong test was TC-36: the assistant asserted that the admin statistics return a 12-month series, which looked plausible and matched a nearby endpoint, but the SRS never states a period, so the "expected value" came from guessing rather than from the requirement. It was caught because the test failed and, before changing code, the expected value was traced back to FR-47, which only asks for "charts and reports". A second, subtler case was TC-65: it passed when it should have tested a rejection, because PHP's array union silently kept the valid category; it was caught by reading the request body the test actually sent. Both show the test-oracle trap from the guideline: a tool reads code and patterns, not requirements, so every expected value had to be checked against the SRS.

11. Test Completion Summary

11.1 Exit-criteria evaluation

Of the 12 Baseline criteria in §3.3 (the branch-coverage row has no Baseline target), 10 are met: 100% of FRs have test cases, all 96 priority-1 cases pass, core-layer line coverage is 92.7% (80.2% for the whole API), no S1 defect is open, the linter reports 0 errors, the dependency audit and secret scan are clean, p95 is measured and below 2 s on all five endpoints, Lighthouse accessibility is ≥ 80 on all six key screens, and the suite runs with one command on a fresh clone. Not met: (1) only one browser engine was tested — Firefox was blocked by the test machine's application-control policy and Safari needs Apple hardware; (2) the usability test with three outside users has not been held yet. Outside the Baseline criteria: one S2 defect (DEF-09, place reviews) stays open by decision, and CI is deferred to Experiment 12.

11.2 Residual risks and known issues

Issue / risk	Impact on users	Likelihood	Mitigation / workaround
Place reviews disabled (DEF-09)	Tourists cannot rate places	Certain	Rate guides and products; decide the visit-record rule and enable
CORS open to any origin (DEF-15)	Other sites can call the public API from a browser	Medium	Set CORS_ORIGIN on Render (config only)
Untested on Firefox / Safari	Layout or map issues on those browsers unknown	Low–Medium	Manual check on a phone (Safari) and Firefox before release
Payments recorded manually	Wrong or fake TrxIDs are not detected automatically	Medium	Admin matches number + TrxID in Payments & Advances against wallet statements
502 for ~1 min after each deploy (DEF-12)	Pages fail to load during deploys	Certain per deploy	Deploy off-peak; paid plan or health-checked rollout
Accessibility issues (DEF-11)	Harder for screen-reader and low-vision users	Certain	Add aria-labels and darker muted text
trips_controller 50.3% covered	Guide-shared trip plans less tested	Low	Add tests for trip ↔ guide responses
Demo accounts with the seed password in production	Account takeover of demo accounts incl. admin	High if public	Change the admin and demo passwords (docs/known-issues.md #7)

11.3 Release recommendation

Go ☐ Go with known issues ☑ — see §11.2 No-Go ☐

HeritageBD can be released to users as a demonstration of the platform with the issues above. All priority-1 test cases pass, including every money and access-control rule, and the one critical defect found (a tourist choosing their own refund) is fixed and guarded by a regression test. Before real customers use it, the team should change the demo/admin passwords, restrict CORS, and run the usability and Firefox/Safari checks; payments must stay clearly labelled as manually verified until a gateway is integrated.

11.4 Lessons learned

- Testing against the SRS rather than against the code found real gaps: cancellation rules, refunds and vendor rejection behaved differently from the specification, and only a requirement-based oracle exposed them.
- Running the API in tests on the production database engine (PostgreSQL, not MySQL) mattered: an earlier MySQL-only literal broke a payment feature in production.
- Tests must create their own data; four tests first passed or failed depending on run order or seed data.
- Next time: write the SRS non-functional targets as numbers (response time, browsers) so they can be tested, and add CI early so every push is checked.

12. Individual Contribution

Member	Test cases designed (IDs)	Tests automated	Defects found	Defects fixed	Test-related PRs
Sumaiya Jahan (0432320005101036)	____	____	____	____	____
Monika Akter (0432320005101047)	____	____	____	____	Commits 99d71bf, 1d45c59

To be completed by the team. Tool assistance for the whole suite is disclosed in §10.

13. Sign-off and Declaration

We declare that this report describes testing carried out on the stated version of our software. All results are reproducible from the repository. All tool assistance is disclosed in §10.

	Name	Signature	Date
Prepared by			
Reviewed by			

Appendix A — Evidence Index

All files are in docs/testing/evidence/ of the repository at the tested version, unless a URL is given.

#	Evidence	File / URL	Supports section
A1	PHP test run log and JUnit (unit + integration)	test-run-coverage.log, junit-php.xml	§6
A2	Frontend and E2E JUnit; Playwright HTML report and screenshots	junit-frontend.xml, junit-e2e.xml, e2e-report/, e2e-screenshots/	§6, §8.4
A3	Generated test catalogue and RTM	test-catalogue.json, rtm.csv	§4, §5
A4	Coverage report (HTML, Clover, summary)	coverage/index.html, coverage-clover.xml, coverage-summary.json, coverage-frontend/	§7
A5	Performance test summary	performance-p95.json, performance-p95.txt	§8.1
A6	Dependency audits and secret scan	npm-audit.json, composer-audit.txt, gitleaks-history.json, gitleaks.txt	§8.2
A7	Lighthouse reports	lighthouse/*.report.html|json	§8.4
A8	E2E runs (incl. 3 consecutive green runs)	test-run-e2e.log	§6.2
A9	Fresh-clone run of the tag	clean-clone-run.log	§2.3, §6.2
A10	Lint output	lint.log	§8.3
A11	Regression before/after runs for fixed defects	regression-before-after.txt	§9
A12	Transport/CORS security check	security-transport.txt	§8.2
A13	CI run (tagged commit)	Not available (no CI in v1.0)	§6.2
A14	SonarCloud quality gate / OWASP ZAP	Planned for v1.1 (Experiment 10)	§8.2, §8.3

Appendix B — Pre-Submission Checklist

- ☑ Tag/commit SHA stated (v1.0.1-test / 1d45c59), and all results come from it
- ☑ Test suite runs from the README commands on a fresh clone (A9)
- ☐ CI run is green on the tagged commit; link included — no CI in v1.0

- ☑ Every FR and NFR appears in the RTM with ≥1 test case
- ☑ Every P1 test case has a specification (§5.2 blocks + test code)
- ☑ Coverage figures match the attached report
- ☑ No open S1 defects; every closed defect has a regression test
- ☑ All tool-assisted tests logged in §10 (whole suite disclosed as tool-assisted)
- ☑ No real personal data or secrets in test fixtures or screenshots
- ☐ Team ID, roles, contribution table and signatures filled in by the team
- ☐ Exported as Team<ID>_Exp9.pdf with the Lab Report Cover as page 1